Big Mountain Resort Pricing Analysis

Problem Statement

Big Mountain Resort recently increased its operating costs by adding a new chair lift. Management wants to determine whether the current ticket price of approximately 81 USD reflects the value of its facilities compared to similar resorts, and whether a price increase is justified without negatively impacting demand.

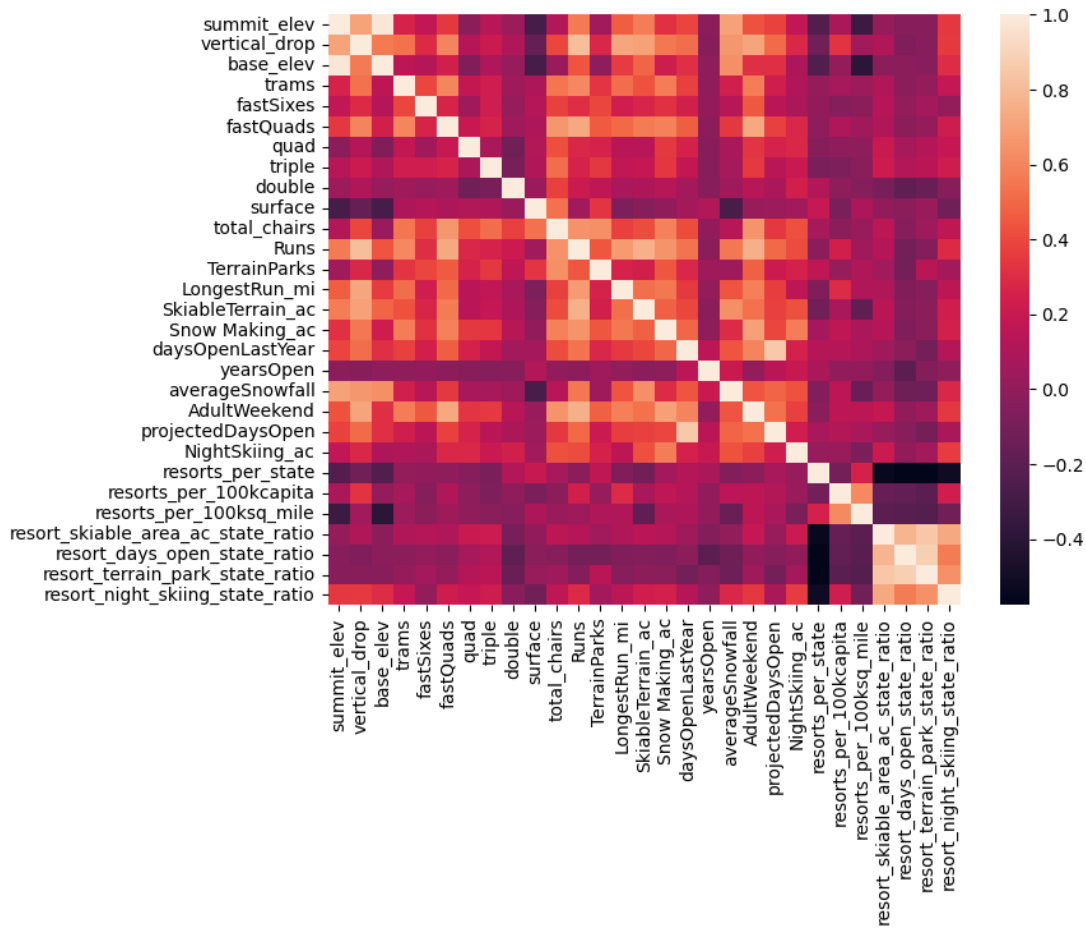
Data Wrangling

The dataset included approximately 330 ski resorts with features such as elevation, number of runs, lift types, terrain, and ticket prices. Missing values were handled by removing incomplete records, and the AdultWeekend ticket price was selected as the target variable due to its consistency. Additional state-level data was merged to enrich the dataset.

Exploratory Data Analysis

Exploratory analysis showed that ticket price is most strongly associated with resort size and infrastructure. The correlation heatmap indicated that features such as vertical drop, number of runs, total chairs, fast quad lifts, and snowmaking area have positive relationships with ticket price.

Scatter plots confirmed that larger resorts with more terrain and better lift infrastructure tend to charge higher prices. These findings suggest that customers value both the size of the resort and the quality of the skiing experience.



Preprocessing, Modeling, and Evaluation

The data was prepared by selecting relevant numerical features and removing redundant variables. Additional ratio-based features were created to better capture relationships between resort characteristics. Categorical variables such as location were not directly included in the model.

Several algorithms were employed, including Linear Regression and Random Forest Regression. Linear Regression was used as a baseline model to capture simple relationships, while Random Forest was used to model more complex, non-linear interactions.

Model performance was evaluated using mean absolute error (MAE). Baseline models produced relatively high errors (around 18 for the mean and around 10 for the median). Linear Regression

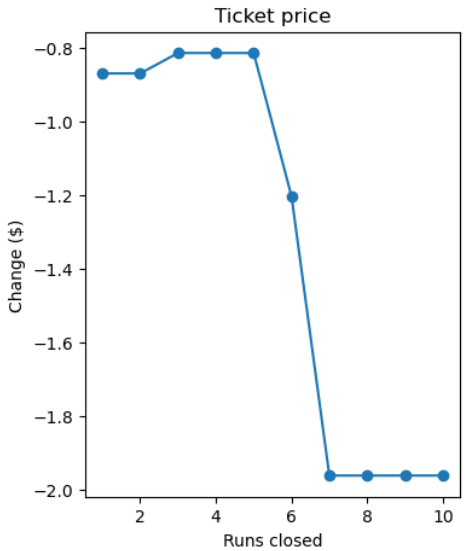
improved performance but remained limited in capturing complexity. Random Forest achieved the best performance, with an MAE of approximately 9.12.

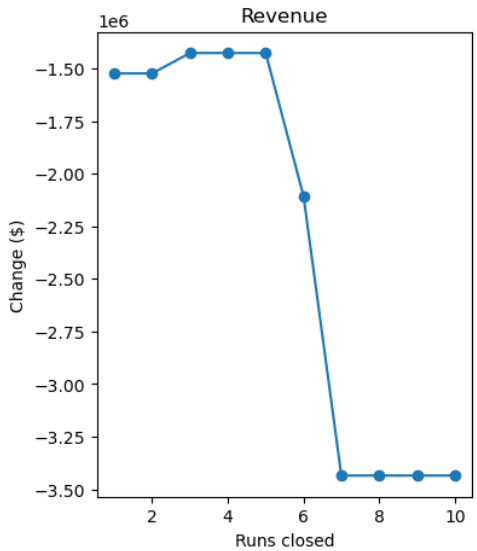
In addition, 5-fold cross-validation was used to ensure that results were consistent across different subsets of the data. Based on these results, Random Forest was selected as the final model.

Feature importance analysis showed that number of runs, vertical drop, and lift infrastructure (such as fast quad lifts) were the most important factors influencing ticket price.

Scenario Modeling

The final model was used to evaluate different operational scenarios. Increasing vertical drop by a run and lift resulted in a small increase in ticket price, which translates into meaningful revenue gains over a full season. Adding additional snowmaking coverage did not significantly affect ticket price. Closing a small number of low-usage runs (1–2) had minimal impact on price and revenue, suggesting a potential opportunity to reduce operating costs. However, closing too many runs (around 6 or more) led to a sharp decline in both ticket price and revenue, indicating that maintaining sufficient terrain is critical for perceived value.





Pricing Recommendation

The model suggests that Big Mountain could increase its ticket price from approximately 81 USD to around 89–90 USD based on its features. However, since the resort is already one of the more expensive options in Montana, any price increase should be gradual and tested carefully to avoid reducing demand.

Conclusion

The analysis shows that ticket price is primarily driven by resort size and lift infrastructure. While small operational improvements can increase revenue, maintaining a sufficient number of runs is essential to preserve customer value.

Future Scope of Work

This analysis is limited by the lack of data on operating costs and customer demand. Future work could include incorporating cost data, customer behavior, and regional competition to better evaluate profitability and pricing strategy.